SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	192521010

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I P.Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Guru

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

ANSWERS FOR THE ABOVE ASSESMENT QUESTION

1) Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

ANS:

Type-1 Hypervisor (Bare-Metal):

- Runs directly on the physical hardware without a host operating system.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen Project.
- Offers high performance, low latency, and better resource utilization.
- Provides stronger security because there is no host OS to attack.
- Supports advanced features such as live migration, clustering, high availability (HA).

Type-2 Hypervisor (Hosted):

- Runs as an application on top of a host operating system.
- Examples: Oracle VM VirtualBox, VMware Workstation.
- Easier to install and use.
- Lower performance due to host OS overhead.
- Higher security risks because both the host OS and hypervisor must be protected.
- Mainly used for testing, development, and desktop virtualization.

Justification: For mission-critical cloud applications, Type-1 hypervisors are the better choice because they provide higher performance, stronger isolation, improved security, and enterprise-grade management features such as live migration, HA, and disaster recovery.

2) **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

ANS:

Virtualization Architecture for Workload Isolation:

- Virtualization allows multiple virtual machines (VMs) to share one physical server while remaining isolated.

How isolation is achieved:

- Each VM has its own virtual CPU, memory, storage, and virtual network.
- The hypervisor controls access to physical resources.
- Memory isolation prevents one VM from reading another VM's memory.
- Network virtualization separates traffic using VLANs or virtual switches.
- Storage virtualization provides separate virtual disks for each workload.
- Thus, finance, healthcare, and student portal applications can safely run on the same hardware without interfering with one another.

Two Major Attack Surfaces:

1. Hypervisor Attacks:

- Attackers exploit vulnerabilities in the hypervisor.
- A compromised hypervisor can expose all hosted VMs.

2. VM Escape:

- Malicious software escapes from a VM to the hypervisor or other VMs.
- Can lead to unauthorized access and data leakage.
- Conclusion: Using secure hypervisors, timely patching, encryption, network segmentation, and access control significantly improves workload isolation.

3) **Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.**

ANS:

VM Host Utilization Analysis:

1) **Given Snapshot:**
   - CPU Utilization = 92%
   - Memory Utilization = 88%
   - Storage I/O Latency = 35 ms
   - Average VM Boot Time = 170 seconds.

2) **Analysis:**
   - CPU (92%)
   - CPU is heavily overloaded.
   - VMs compete for processor time.
   - Applications may respond slowly.

   Corrective Action:
   - Migrate VMs to another host.
   - Increase CPU cores.
   - Apply CPU resource scheduling.

3) **Memory (88%):**
   - Memory usage is very high.
   - System may start swapping to disk.

   Corrective Action:
   - Add more RAM.
   - Reduce VM memory allocation.
   - Use memory optimization techniques.

4) **Storage Latency (35 ms):**

- Storage latency is much higher than the recommended (<10 ms).
- Disk operations become slow.

Corrective Action:

- Upgrade to SSD/NVMe storage.
- Improve storage controller performance.
- **Balance storage workloads.**

5) **VM Boot Time (170 s):**

- Boot time is unusually high.
- Caused mainly by CPU overload and storage bottlenecks.

Corrective Action:

- Optimize VM templates.
- Improve storage speed.
- Reduce simultaneous VM startups.

4) **Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.**
ANS:

Software Defined Datacenter (SDDC) vs Traditional Datacenter:

- A Software Defined Datacenter (SDDC) virtualizes compute, storage, and networking, enabling centralized software-based management.

Traditional Datacenter:

- Manuel configuration
- Hardware dependent
- Slow deployment
- Limited scalability
- Complex management

Software defined Datacentre:

- Automated provisioning
- Software controlling
- Rapid deployment
- High scalability
- Centralized management

Compute Virtualization:

- Creates virtual machines from physical servers.
- Enables live migration and automatic load balancing.

Storage Virtualization:

- Pools storage devices into one logical storage system.
- Simplifies expansion and improves utilization.

Network Virtualization:

- Creates virtual networks independent of hardware.
- Enables rapid network provisioning and security policies.

5) **A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.**

ANS:

Identity Federation in Multi-Cloud Environments:

Problem:

- A multi-cloud federation project fails because each cloud provider maintains separate user identities and authentication systems.

This causes:

- Duplicate user accounts.
- Authentication failures.
- Poor user experience.
- Security inconsistencies.

Secure Identity and Privacy Design:

Identity Federation:

- Allow users to authenticate once using a trusted Identity Provider (IdP).
- All cloud providers trust the same identity.

Single Sign-On (SSO):

- Users log in once and securely access multiple cloud services.

Standard Federation Protocols:

- Use SAML 2.0, OAuth 2.0, or OpenID Connect for secure identity exchange.

Privacy Protection:

- Share only the minimum user information required.
- Encrypt identity tokens.
- Apply Multi-Factor Authentication (MFA).
- Use Role-Based Access Control (RBAC).
- Maintain audit logs for monitoring.

Advantages:

- Centralized identity management.
- Better security.
- Reduced password management.
- Consistent user experience.
- Improved privacy and compliance.